

Level-Aware AI Coaching for Human Skill Development

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Abstract—AI copilots can substantially boost human performance through shared control, but excessive assistance can induce over-reliance and skill atrophy. This paper studies how an embodied AI agent can act as a coach that accelerates human motor-skill development. We argue that effective coaching requires strategic scaffolding and stepping back that are aligned with the learner’s capability, allowing productive failures that drive learning. We formalize the interactive AI coaching process as a non-cooperative dynamic game in which the learner optimizes task performance while the coach targets the learner’s independent competence. Building on this formalism, we develop a reinforcement learning framework combining adaptive shared control with probabilistic models of the coach’s causal influence on skill evolution. A user study ($N = 33$) on first-person-view drone racing shows significant gains in human learning outcomes over state-of-the-art AI coaching baselines. We use observed task behavior, including segment passage time, to estimate latent skill and adapt the coaching strategy.

I. Introduction

Mastering a motor skill, from racing a car at the limit, to piloting a drone through tight gates, requires deliberate practice at the boundary of one’s current ability. Human coaches accelerate this process not by performing the task for the learner, but by orchestrating the right challenges at the right moments [1], [2]. Can an embodied AI agent assume this role?

Recent advances in robot learning have produced embodied AI agents that can outperform world champions [3], [4]. A natural use of such expert AI is as a shared-control copilot: a system that fuses the human’s commands with the AI’s corrective inputs to keep the team performing well despite human error [5], [6], [7], [8], [9]. This is the dominant paradigm in modern human–AI collaboration, where AI acts as a “guardian angel” that prevents failures and smooths execution. However, helping humans succeed now is fundamentally different from helping them succeed alone in the future. Persistent AI assistance can induce over-reliance and skill stagnation in human users [10], [11], depriving them of the productive failures [1] that drive skill development and retention. Conversely, a coach that withdraws assistance indiscriminately exposes the learner to failures that are uninformative or frustrating, neither of which benefits human skill learning.

Our key insight is that an effective AI coach must neither overprotect nor disengage from the learner; it should strategically step back when doing so turns mistakes

into catalysts of long-term independent competence. We formalize this through the Human–AI Coaching Game, a non-cooperative dynamic game in which a learner optimizes task performance while a coach targets the learner’s long-term independent competence. We develop a reinforcement learning framework combining an adaptive, closed-loop shared control rule, a probabilistic automaton that captures the coach’s causal influence on learner skill, and a tractable skill-improvement reward for scalable coaching policy training. We deploy our AI coach in a user study ($N = 33$) of first-person-view (FPV) drone racing with a high-fidelity simulator, demonstrating significant gains in human learning over state-of-the-art AI coaches.

Since skill is unobservable at deployment, the coach infers it from task behavior. In our drone-racing setting, the observation is the realized passage time for each gate segment. We use this simple cue to ask how capability inference should be evaluated when the robot also changes the observed trajectory.

II. The Human–AI Coaching Game

At each moment, the coach must decide how much it scaffolds the learner toward task completion, or steps back to let them struggle, fail, and learn. Intervening aggressively risks coddling, leaving the learner over-reliant on AI; withdrawing too early yields uninformative failures and frustration. Moreover, the learner’s true skill is never directly observed and must be inferred from behavior as it evolves.

We cast the AI coaching problem as a two-player partially observable stochastic game. The learner seeks to maximize task performance, while the coach optimizes a pedagogical objective, namely the learner’s independent competence. These distinct incentives are critical for coaching success: the coach may modulate assistance when doing so accelerates long-term human skill development, even at the cost of short-term task performance. Intuitively, the coach’s Value of Independence asks: “If the coach were to walk away right now, how well would the learner perform?”

The AI coach implements shared control by fusing the learner’s action with an expert baseline via a linear blending rule:

$$a_t = \lambda_t \odot \pi^*(o_t^C) + (\mathbf{1} - \lambda_t) \odot a_t^H, \quad (1)$$

where $\lambda_t \in [0, 1]^{\dim(\mathcal{A})}$ is a per-axis blending vector that modulates the assistance level, from zero (no assistance) to one (full override), and π^* is a pretrained expert

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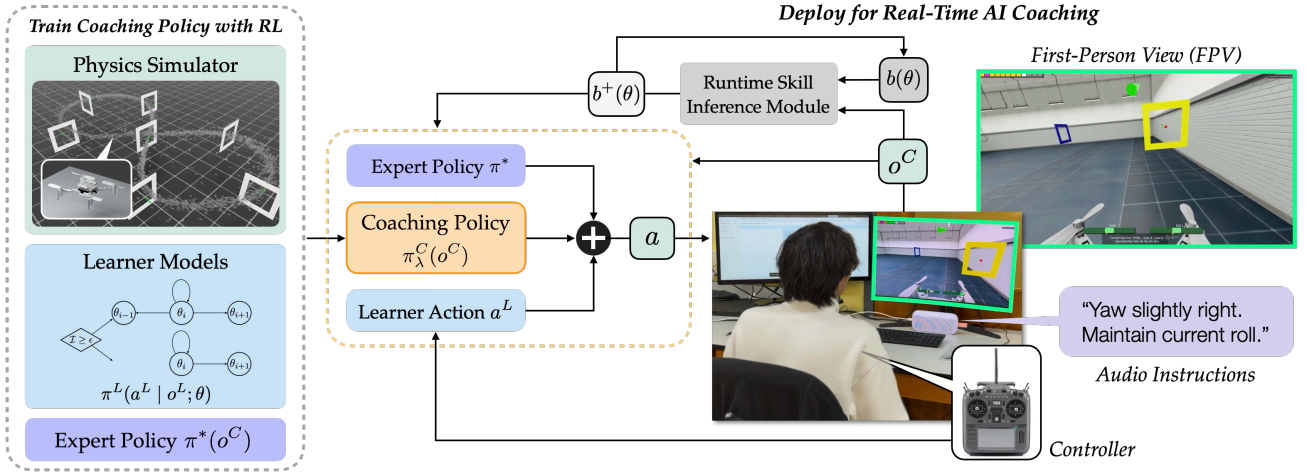


Fig. 1. Our AI coach accelerates human motor-skill development through strategic scaffolding and stepping back. The coach adaptively blends the expert action with the learner’s command. At deployment, the AI maintains a belief over the learner’s latent skill and conveys feedback through shared control, audio instructions, and visual cues.

policy. Anchoring on a pretrained expert baseline decouples task competence from coaching strategy: the coach only learns to modulate assistance rather than synthesize full control from scratch. Unlike fixed modulation rules, our blending factor is observation-dependent and adapts in real time to both the physical environment and the learner’s action.

III. Learning to Coach

To train a coaching policy, we augment the robot’s transition dynamics with two learner-side components: a skill-conditioned control policy, and a model of how the learner’s latent skill θ evolves in response to coaching events such as successful or failed task attempts. Given an expert value function $Q^*(o^H, a)$, we model learners of different skill as Boltzmann noisily-rational decision-makers whose action variance shrinks with proficiency:

$$\pi^H(a^H | o^H; \theta) \propto \exp(\beta(\theta)Q^*(o^H, a^H)), \quad (2)$$

where $\beta(\theta)$ increases monotonically in θ . A simulated novice samples actions broadly, while more skilled learners concentrate their actions more sharply around the expert action. Our framework is designed to be modular with respect to the learner model: it only requires a θ -parameterized policy class. The Boltzmann model is one tractable choice that is simple and interpretable.

We model the learner’s skill change as probabilistic transitions triggered by success and failure events. The learner is deemed to have failed when a task-level performance index falls below a prescribed threshold. When the learner succeeds, they upskill, downskill, or remain unchanged with skill-dependent probabilities; when the learner fails, they may still upskill. The transition rates capture pedagogically meaningful trends: small wins matter most early on, skill becomes harder to lose once consolidated, and skilled learners extract more from their mistakes. To cover a diverse learner population,

we randomize the transition parameters at the start of each training episode.

We train the coaching policy with proximal policy optimization, leveraging privileged access to the learner’s skill level in simulation. The coach receives the tractable surrogate reward for skill improvement

$$r_t^C = \theta_t - \theta_{t-1}. \quad (3)$$

Under the assumption that independent task value is nondecreasing in skill, a policy that induces a stochastically higher skill trajectory also obtains a higher discounted independent-value return. At deployment, skill is unobservable. The coach maintains a belief b_t over θ and updates it through runtime inference from observed learner behavior.

In drone racing, we partition the track into per-gate segments and maintain an independent belief over the learner’s skill at each gate. The per-gate likelihood is a Gaussian on realized segment passage time, centered at a skill-dependent target time calibrated from a pilot study.

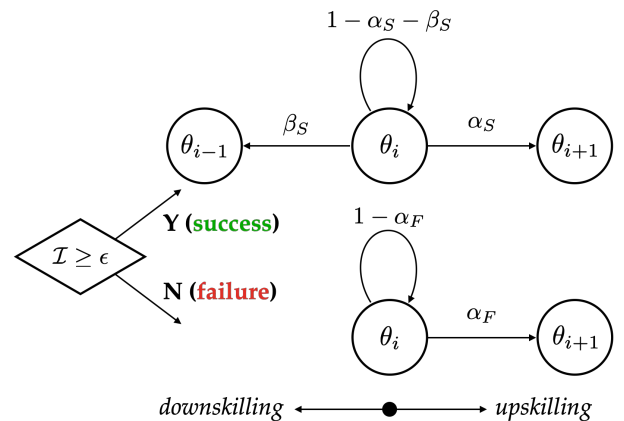


Fig. 2. Hybrid probabilistic finite-state automaton for simulating skill change triggered by success or failure events.

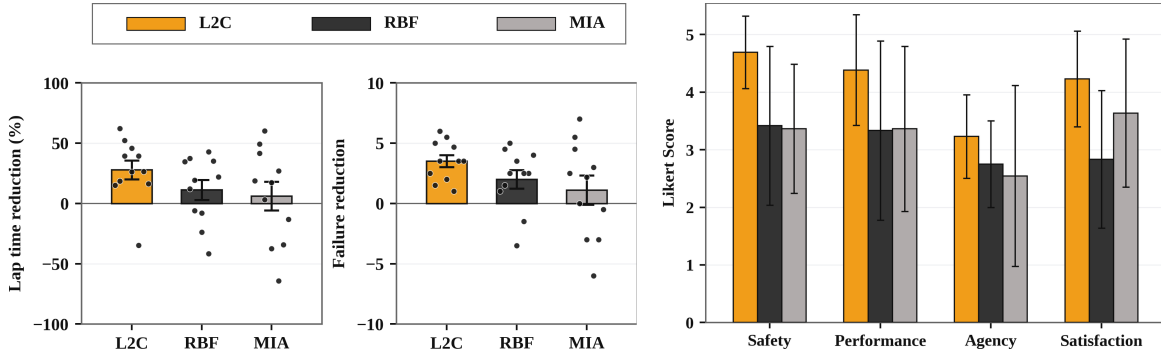


Fig. 3. After coaching, learners trained with L2C show significant reductions in lap time and failure count and report higher safety, performance, agency, and satisfaction than both baselines.

Faster passages shift belief mass toward higher skill, slower passages toward lower. Upon failure, the learner restarts one gate back and re-attempts the failed segment with targeted guidance from the coaching system.

IV. User Study

We conducted a user study with $N = 33$ participants from diverse drone-flying backgrounds in a high-fidelity FPV drone-racing simulator. Each trial has three phases: two unassisted pre-test laps, 15 coached training laps (around 40 minutes), and two unassisted post-test laps. Participants were randomly assigned to L2C, rule-based fading (RBF) [12], or minimally invasive assistance (MIA). All three methods use the same simulator, track, expert policy, language instructions, and visual cues.

For L2C, a paired-samples test yielded a mean lap-time reduction of 27.9% with $p = 0.005$ and Cohen’s $d_z = -1.08$. The mean failure count decreased by 3.52 failures per lap with $p < 0.001$ and $d_z = -2.13$. RBF reduced lap time by 11.3% and failures by 2.00 per lap; MIA reduced lap time by 6.2% and failures by 1.11. L2C is the only method to reach within-subject significance on both outcomes. The learned intervention is not a global fading schedule: assistance changes with estimated skill and varies sharply around gates that are critical to the task.

V. Task Behavior as a Cue for Skill

Since θ is unobservable at deployment, the coach maintains a belief b_t over skill and updates it through Bayesian inference,

$$b^+(\theta) \propto P(o^C | \theta)b(\theta). \quad (4)$$

In drone racing, the track is partitioned into per-gate segments and an independent belief is maintained for each gate. The likelihood is a Gaussian on realized passage time, centered at a skill-dependent target time calibrated from a pilot study. Under this model, faster passages shift belief mass toward higher skill and slower passages toward lower skill.

Passage time is available from the simulator without a camera, wearable, or verbal report. It is also measured at the same task segment on repeated laps. These properties

make it a useful first signal for capability inference. The claim is not that passage time identifies every human state. It may also reflect hesitation, strategy, fatigue, or perception. The belief only represents the part of behavior that the current model calls skill.

Skill also changes over time. The probabilistic automaton updates it after success and failure, while the Bayesian observation model updates the coach’s belief about its current value. A single slow passage does not determine the label: evidence accumulates through interaction, and gates can retain distinct beliefs.

VI. The Coach Changes the Observation

An important complication is that the coach acts on the same trajectory from which it infers skill. The executed action combines the learner’s command and the expert action. A fast passage with λ near one is therefore not direct evidence that the learner can repeat the passage alone. Treating assisted success as independent competence would reward the coach for making its own observation easier.

Our study addresses this problem with unassisted evaluation laps. Every third coached lap is unassisted, providing a clean signal for updating the per-gate belief. The pre- and post-coaching tests are also unassisted. These laps correspond to the evaluation distribution used in the Value of Independence: they measure how the learner performs when the coach walks away.

The logged human and expert actions provide another useful signal. Their residual shows how much correction the expert proposed before blending. Passage time together with this residual can distinguish two traversals with similar executed performance but different human contributions. A richer observation model can condition its likelihood on both the segment outcome and the assistance magnitude.

VII. A Cue Evaluation Protocol

A cue model for coaching should first be tested on future unassisted behavior. For each gate, the history of passage time, failures, human commands, and assistance can be used to predict the next unassisted passage time and failure event. Models can be compared by predictive

likelihood and calibration, not only by classification accuracy on assisted trajectories.

The probe schedule should be fixed across methods. Otherwise, a coach that asks for a probe only at easy gates changes the evaluation distribution. The same initial-skill distribution, task segments, and assistance-policy architecture should be used when cue sets are compared. This isolates whether an additional cue improves inference rather than giving the policy a different task.

The second test is whether the cue changes assistance in a useful way. One comparison can freeze the coaching policy and replace only the inferred belief. Another can retrain the same policy architecture for each cue set. In both cases, assistance magnitude, failure exposure, and post-coaching unassisted performance should be reported together. A cue that only produces more expert override may improve assisted completion without improving learning.

Calibration should be stratified by gate and assistance level. If confidence grows after heavily assisted passages but does not predict the next evaluation lap, the model is assigning the coach’s contribution to the learner. Overestimating skill may expose a novice to failures; underestimating skill may coddle a capable learner. Evaluations should distinguish the two errors.

VIII. Adding Other Nonverbal Signals

The framework does not require passage time. Alternative skill-inference schemes include Bayesian Knowledge Tracing, although those methods often require offline fitting from large human datasets. Control trajectories, gaze, or verbal feedback can also update the belief through learned likelihoods.

The lowest-burden comparison is telemetry alone. Human-expert action residuals can test whether disagreement predicts the need for help, while gaze, speech, or physiology may separate confusion from fatigue. Added signals should be judged by prediction on unassisted probes and subsequent independent performance, not an offline cue-recognition benchmark. Missing observations should widen the belief or trigger fallback to telemetry rather than count as evidence of low skill.

Beyond drone racing, the segment can be replaced by a maneuver in driving, manipulation, rehabilitation, or teleoperation. The observation model and probe change with the task, but the evaluation remains the same: does the cue help the robot choose assistance that leaves the person more capable when assistance is removed?

Nonverbal signals differ in reliability: telemetry is continuous, gaze may be intermittent, and speech or physiology delayed. Benchmarks should test controlled cue dropout, latency, and corruption. The coach should fall back to telemetry rather than interpret missing data as low skill; with no reliable cue, it should express uncertainty and bound assistance.

IX. Relation to Learner Modeling

The framework only requires a θ -parameterized learner policy. The Boltzmann model is one tractable choice, but behavior cloning, fictitious co-play, dynamic games, inverse games, or mixtures of policies can replace it without redesigning the coaching algorithm. Runtime inference is similarly modular: our passage-time likelihood is deliberately simple and updated online, whereas alternatives such as Bayesian Knowledge Tracing generally require offline fitting from larger human datasets.

X. Observed Behavior under Coaching

The learned blending vector shows two forms of adaptation. Across learners, the novice estimate receives more assistance than the higher-skill estimate. Within trajectories, assistance changes around task-critical gates. A fixed fading schedule does not produce these patterns.

The user study measures skill change with lap time and failure count. L2C is the only method with significant within-subject improvement in both outcomes. On the post-trial questionnaire, participants also reported higher safety, performance, agency, and satisfaction. The agency item is reverse-coded from the statement that the AI coach intervened too much. Taken together with the assistance maps, these outcomes show that the inferred skill is used to change intervention without simply increasing expert authority everywhere.

Richer cues should be tested under the same assistance rule and unassisted evaluation laps, because better offline prediction need not produce better coaching. The useful cue is the one that helps the coach decide when to scaffold and when to step back, yielding higher independent competence after assistance is removed.

XI. Limitations and Outlook

Although our probabilistic skill model appeared sufficient for drone racing, the simulated learners may not capture the full variability of real human learning in more complex motor-skill training settings. Future work could explore data-driven alternatives such as behavior cloning, world models, or mixtures of policies. Our current system learns only the assistance modulation of physical control, while verbal and visual cues are predefined and triggered by fixed rules. Future work could jointly optimize a multimodal coaching policy with a foundation-model backbone.

XII. Conclusion

As AI systems grow more capable, the question is no longer whether they can replace human skill, but how they can help build it. We have argued, and shown empirically, that optimizing for independent human competence gives both a principled definition of human-empowering AI and a practical recipe for building one. In a comprehensive drone-racing user study, our game-informed RL coach measurably accelerates the development of human skill.

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